

🔥 XAI: Understanding Course Foundations and Motivations

FAMAF - UNC

September 9, 2025



**eXplainable
Artificial Intelligence**

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Agenda ---

- Course Goals & Logistics
- Model Understanding: Use Cases
- Inherently Interpretable Models vs. Post-hoc Explanations
- Defining and Understanding Interpretability

🍷 Goals of this Course ---

Learn and improve upon the state-of-the-art literature on ML interpretability and explainability:

- Understand where, when, and why is interpretability/explainability needed
- Read, present, and discuss research papers
- Formulate, optimize, and evaluate algorithms
- Implement state-of-the-art algorithms; Do research!
- Understand, critique, and redefine literature

EMERGING FIELD!!

Course Overview ---

- ① **Foundations and Human Factors** - definitions, taxonomies, and cognitive aspects
- ② **Interpretability and Explainability Methods** - inherently interpretable models and post-hoc explanations
- ③ **Advanced Techniques and Evaluation** - attention-based explanations and quality metrics
- ④ **Interpretability of Large Models** - mechanistic interpretability and LLM understanding
- ⑤ **Emerging Frontiers of XAI** - automated circuit discovery and multimodal techniques

🍷 Who should take this course? ---

- Course particularly tailored to students interested in **research** 📝 on interpretability/explainability
 - ▶ **Not a surface level course!** 🔥
 - ▶ **Not just applications!** 🔥
- Goal is to push you to question existing work and make new contributions to the field

Class Format ---

- Course comprises of lectures, guests, and student presentations
- Each lecture will cover: at least 2 papers  
- Students are **expected** to “at least” skim through the papers beforehand 
- Students will divide into groups

Each breakout group is expected to come up with:

- A list of 2 to 3 weaknesses of each of the works discussed
- Strategies for addressing those weaknesses

Research Projects - AKA Final Exam ---

Requirements

- Short research paper (max 4 pages)
- Teams of 2 students
- Target: Local conference submission

Assessment

- Ongoing feedback from entire course
- Course approved when paper is submitted

Background ---

Understanding:

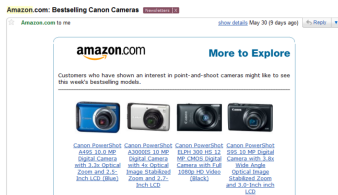
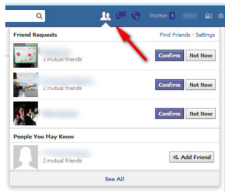
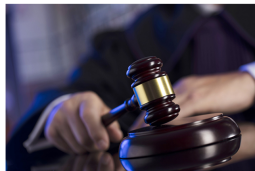
- Linear algebra
- Probability
- Algorithms
- Machine learning
- Programming in Python, Numpy, Sklearn

Familiarity with:

- Statistics
- Optimization

🍎 Motivation ---

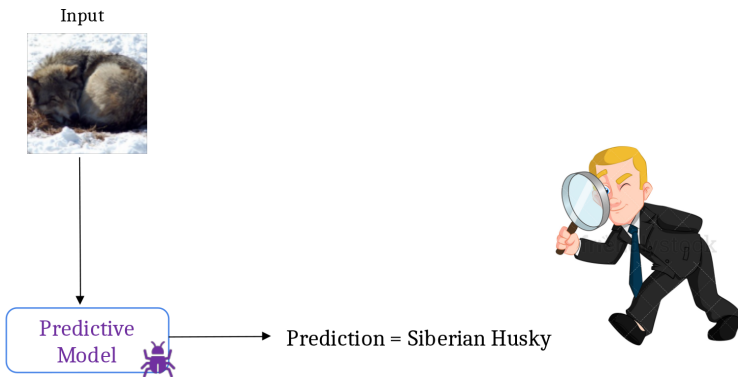
Machine Learning is EVERYWHERE!!



(Lipton 2016)

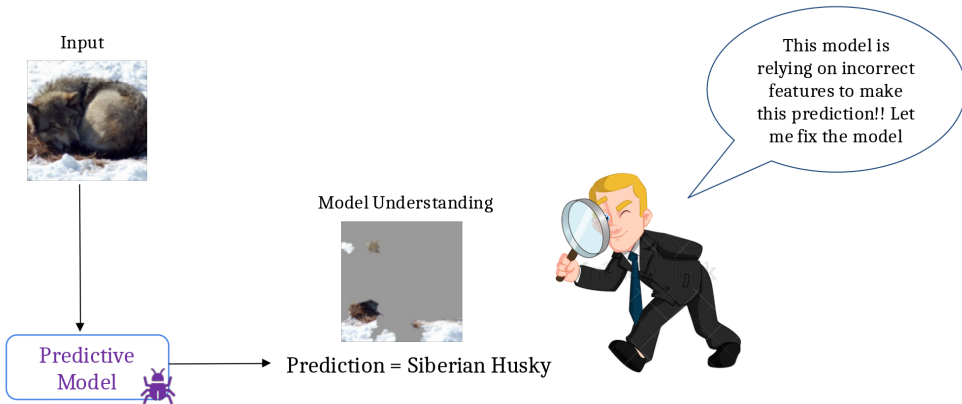
🍷 Motivation: Why Model Understanding? ---

Example: Image classification model



🍷 Motivation: Why Model Understanding? ---

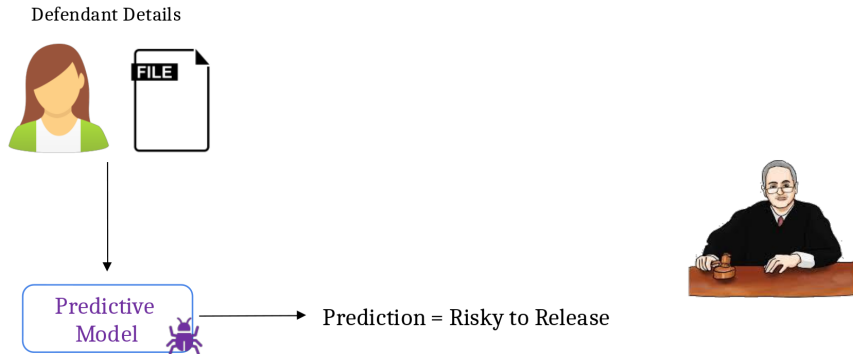
Example: Image classification model



Model understanding facilitates debugging

🍷 Motivation: Why Model Understanding? ---

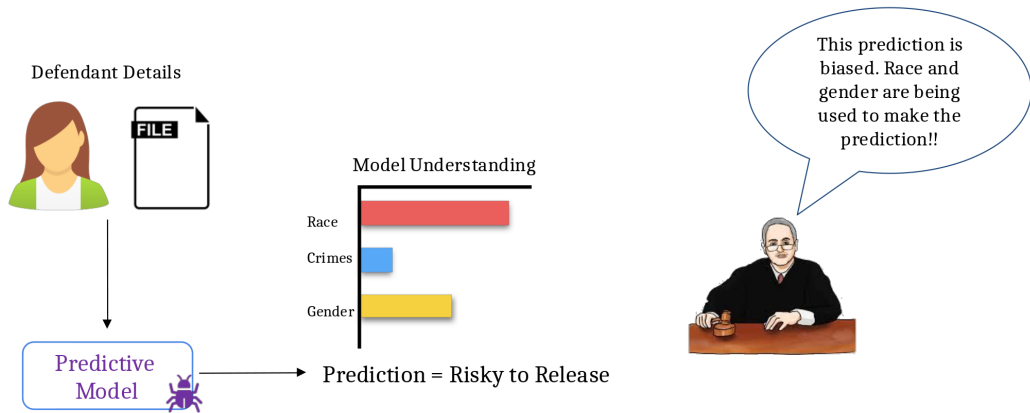
Example: Criminal justice risk assessment



(Lipton 2016)

🍷 Motivation: Why Model Understanding? ---

Example: Criminal justice risk assessment



Model understanding facilitates bias detection

(Lipton 2016)

Motivation: Why Model Understanding? ---

Example: Loan application system

Loan Applicant Details



Predictive
Model



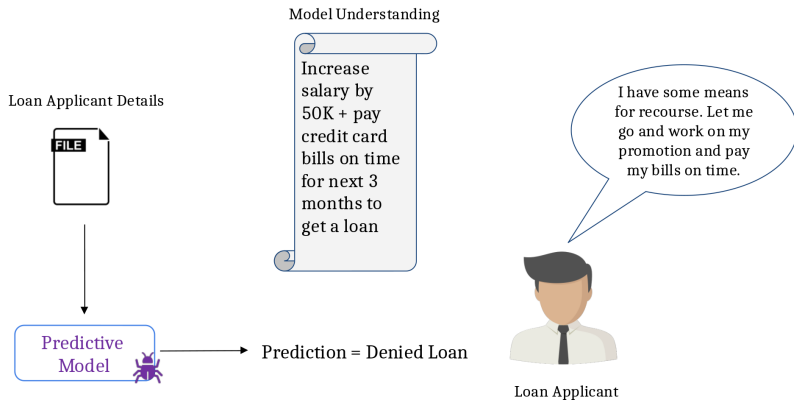
Prediction = Denied Loan



Loan Applicant

🍷 Motivation: Why Model Understanding? ---

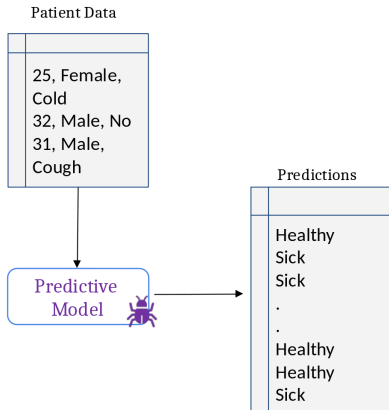
Example: Loan application system



Model understanding helps provide recourse to individuals who are adversely affected by model predictions

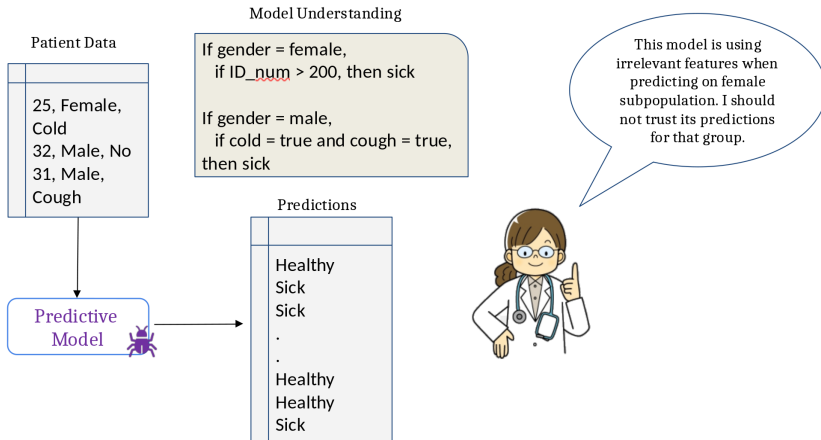
🍷 Motivation: Why Model Understanding? ---

Example: Medical diagnosis system



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Example: Medical diagnosis system



Model understanding helps assess when to trust predictions

Motivation: Why Model Understanding? ---

Summary of Use Cases

Utility

Debugging

Bias Detection

Recourse

If and when to trust model predictions

Vet models to assess suitability for deployment

Motivation: Why Model Understanding? ---

Summary of Use Cases

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Recourse

If and when to trust model predictions

Vet models to assess suitability for deployment

Stakeholders

End users (e.g., loan applicants)

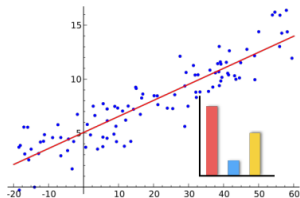
Decision makers (e.g., doctors, judges)

Regulatory agencies (e.g., FDA, European commission)

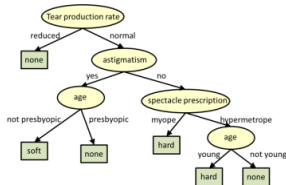
Researchers and engineers

🍵 Achieving Model Understanding ---

Take 1: Build inherently interpretable predictive models



Linear regression



Decision trees

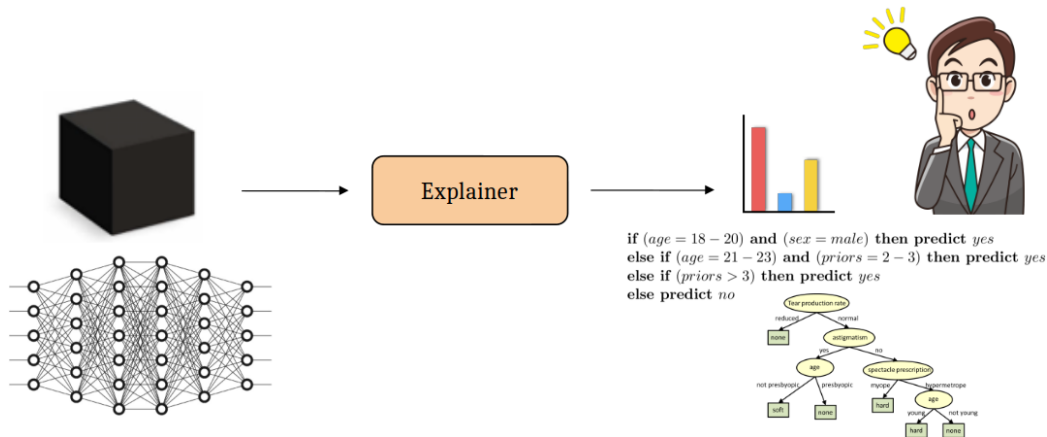
if (*age* = 18 – 20) and (*sex* = *male*) then predict *yes*
else if (*age* = 21 – 23) and (*priors* = 2 – 3) then predict *yes*
else if (*priors* > 3) then predict *yes*
else predict *no*

Rule-based models

(Letham et al. 2015; Lakkaraju, Bach, and Leskovec 2016)

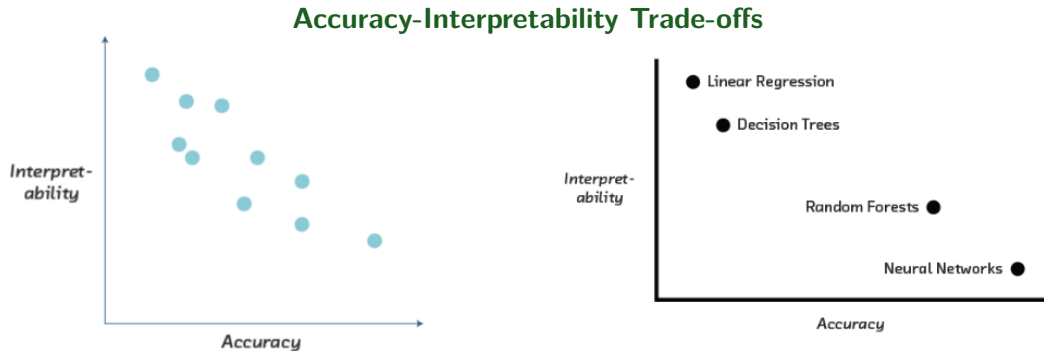
🧠 Achieving Model Understanding ---

Take 2: Explain pre-built models in a post-hoc manner



(Ribeiro, Singh, and Guestrin 2016, 2018; Lakkaraju and Lage 2019)

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

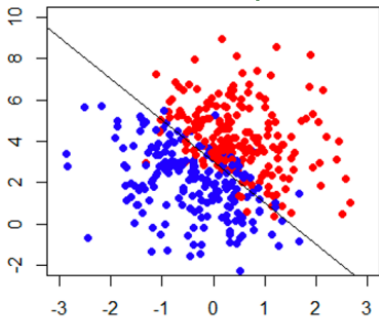


In certain settings, accuracy-interpretability trade offs may exist.

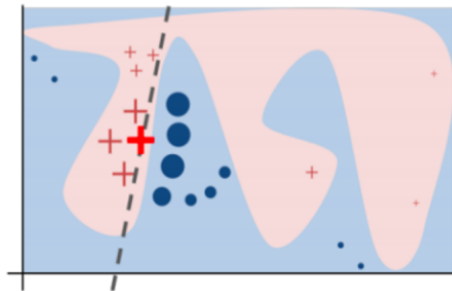
(Cireřan et. al. 2012, Caruana et. al. 2006, Frosst et. al. 2017, Stewart 2020)

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

Example scenarios: Simple vs complex boundaries



can build interpretable + accurate models



complex models might achieve higher accuracy

In certain settings, accuracy-interpretability trade offs may exist.

🍷 Inherently Interpretable Models vs. Post hoc Explanations ---

Sometimes, you don't have enough data to build your model from scratch. And, all you have is a (proprietary) black box!

(Ribeiro, Singh, and Guestrin 2016)

🍲 Inherently Interpretable Models vs. Post hoc Explanations ---

Recommendation

If you can build an interpretable model which is also adequately accurate for your setting,

DO IT!

*Otherwise, **post hoc explanations** come to the rescue!*

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Let's get into some details!

🍷 Next Up! ---

- Define and evaluate interpretability somewhat! 🍷
- Taxonomy of interpretability evaluation
- Taxonomy of interpretability based on applications/tasks
- Taxonomy of interpretability based on methods

🍷 Defining and Understanding Interpretability: Motivation for Interpretability ---

- ML systems are being deployed in complex **high-stakes settings**
- Accuracy alone is no longer enough
- **Auxiliary criteria are important:**
 - ▶ Safety
 - ▶ Nondiscrimination
 - ▶ Right to explanation

🍷 Motivation for Interpretability (cont.) ---

- Auxiliary criteria are often **hard to quantify** (completely):
 - ▶ E.g.: Impossible to enumerate all scenarios violating safety of an autonomous car

Fallback option: *Interpretability*

If the system can explain its reasoning, we can verify if that reasoning is sound w.r.t. auxiliary criteria

🍲 Prior Work: Defining and Measuring Interpretability ---

Bad News

Little consensus on what interpretability is and how to evaluate it.

Interpretability evaluation typically falls into:

① Evaluate in the context of an application

- ▶ If a system is useful in a practical application or a simplified version, it must be interpretable

② Evaluate via a quantifiable proxy

- ▶ Claim some model class is interpretable and present algorithms to optimize within that class
- ▶ E.g. rule lists

"You will know it when you see it!"

🍷 Lack of Rigor? ---

Yes and No

Previous notions are reasonable

- **However:**

- ▶ Are all models in all “interpretable” model classes equally interpretable?
 - ★ Model sparsity allows for comparison
- ▶ How to compare a linear model with a decision tree?
- ▶ Do all applications have same interpretability needs?

Important to formalize these notions!!!

🍷 What is Interpretability? ---

Definition

Ability to explain or to present in understandable terms to a human

No clear answers in psychology to:

- What constitutes an explanation?
- What makes some explanations better than the others?
- When are explanations sought?

🍷 When and Why Interpretability? ---

Not all ML systems require interpretability

- E.g., ad servers, postal code sorting
- No human intervention
- **No explanation needed because:**
 - ▶ No consequences for unacceptable results
 - ▶ Problem is well studied and validated well in real-world applications → trust system's decision

When do we need explanation then?

🍷 When and Why Interpretability? ---

- **Incompleteness in problem formalization**
 - ▶ Hinders optimization and evaluation
- **Incompleteness $\neq$ Uncertainty**
 - ▶ Uncertainty can be quantified
 - ▶ E.g., trying to learn from a small dataset (uncertainty)

🍷 Incompleteness: Illustrative Examples ---

Scientific Knowledge

- E.g., understanding the characteristics of a large dataset
- Goal is abstract

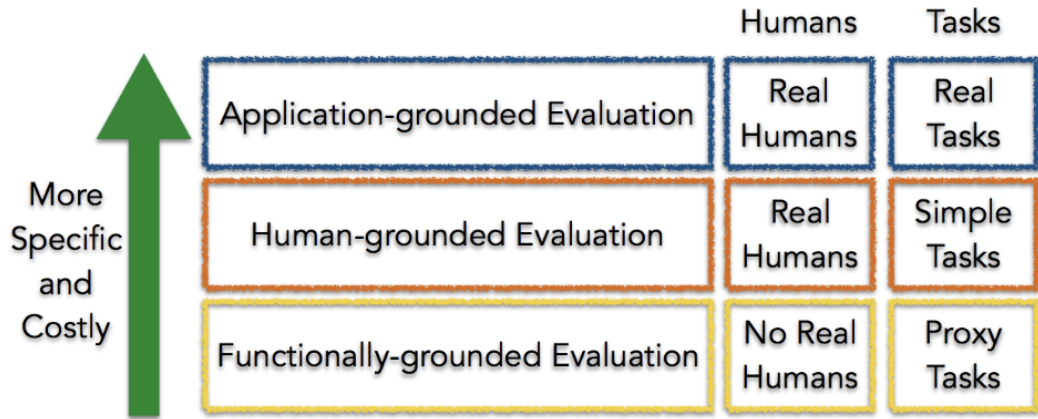
Safety

- End to end system is never completely testable
- Not possible to check all possible inputs

Ethics

- Guard against certain kinds of discrimination which are too abstract to be encoded
- No idea about the nature of discrimination beforehand

🍷 Taxonomy of Interpretability Evaluation ---



Claim of the research should match the type of the evaluation!

🍷 Application-grounded evaluation ---

- Real humans (domain experts), real tasks
- Domain experts experiment with **exact application task**
- Domain experts experiment with a **simpler or partial task**
 - ▶ Shorten experiment time
 - ▶ Increases number of potential subjects
- Typical in *Human-Computer Interaction* (HCI) and visualization communities

🍷 Human-grounded evaluation ---

Real humans, simplified tasks

- Can be completed with lay humans
- Larger pool, less expensive

Potential experiments:

- Pairwise comparisons
- Simulate the model output
- What changes should be made to input to change the output?

🍷 Functionally-grounded evaluation ---

No humans, just proxies

- Appropriate for a class of models already validated (E.g., decision trees)
- A method is not yet mature
- Human subject experiments are unethical
- What proxies to use?

Potential experiments:

- Complexity (of a decision tree) compared to other models of the same (similar) class
 - ▶ How many levels? How many rules?

🍲 Open Problems: Design Issues ---

- What proxies are best for what real world applications? 🙌
- What factors to consider when designing simpler tasks in place of real world tasks? 🙌

Taxonomy based on applications/tasks ---

Global vs. Local

- High level patterns vs. specific decisions

Degree of Incompleteness

- What part of the problem is incomplete? How incomplete is it?
- Incomplete inputs or constraints or costs?

Time Constraints

- How much time can the user spend to understand explanation?

🍷 Taxonomy based on applications/tasks (cont.) ---

Nature of User Expertise

- How experienced is end user?
- Experience affects how users process information
- E.g., domain experts can handle detailed, complex explanations compared to opaque, smaller ones

Note:

These taxonomies are constructed based on intuition and are not data or evidence driven. They must be treated as hypotheses.

Taxonomy based on methods ---

Basic units of explanation:

- Raw features? E.g., pixel values
- Semantically meaningful? E.g., objects in an image
- Prototypes?

Number of basic units of explanation:

- How many does the explanation contain?
- How do various types of basic units interact?
- E.g., prototype vs. feature

Taxonomy based on methods (cont.) ---

Level of compositionality:

- Are the basic units organized in a structured way?
- How do the basic units compose to form higher order units?

Interactions between basic units:

- Combined in linear or non-linear ways?
- Are some combinations easier to understand?

Uncertainty:

- What kind of uncertainty is captured by the methods?
- How easy is it for humans to process uncertainty?

🍷 Relevant Conferences to Explore ---

ML/AI Venues:

- ICML, NeurIPS, ICLR
- UAI, AISTATS, KDD, AAAI

Ethics/HCI Venues:

- FAccT, AIES
- CHI, CSCW, HCOMP

Closing ---

- Say hi to your neighbors! Introduce yourselves!
- What topics are you most excited about learning as part of this course?
- Are you convinced that model interpretability/explainability is important?
- Do you think we can really interpret/explain models (correctly)?
- What is your take on inherently interpretable models vs. post hoc explanations? Would you favor one over the other? Why?

- Lakkaraju, Himabindu, Stephen H Bach, and Jure Leskovec. 2016. "Interpretable Decision Sets: A Joint Framework for Description and Prediction." In *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, 1675–84.
- Lakkaraju, Himabindu, and Ike Lage. 2019. "Interpretability and Explainability in Machine Learning."
- Letham, Benjamin, Cynthia Rudin, Tyler H McCormick, and David Madigan. 2015. "Interpretable Classifiers Using Rules and Bayesian Analysis: Building a Better Stroke Prediction Model."
- Lipton, Zachary C. 2016. "The Mythos of Model Interpretability." *Queue* 14 (3): 31–57.
- Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. 2016. "'Why Should I Trust You?' Explaining the Predictions of Any Classifier." In *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, 1135–44.
- . 2018. "Anchors: High-Precision Model-Agnostic Explanations." In *Proceedings of the Aai Conference on Artificial Intelligence*. Vol. 32. 1.